

6.4 CALCULATING LONGITUDINAL CG (see Fig. 6-2)

The following examples show a method of calculating the longitudinal center of gravity.

NOTE Basic empty mass data are to be taken from the latest entry on the MASS AND BALANCE RECORD, Form MBR-1 appended to this Section.

LOADING EXAMPLES						
Flight A: 1 pilot (80 kg), bulk cargo load (320 kg), and full fuel (456 kg). The resultant CG is solved as follows:						
	Mass (kg)	Arm (mm)	Massmoment (kgmm)	Resultant		
	Mass (kg)	Arm (mm)	Massmoment (kgmm)	Mass (kg)	Arm (mm)	Massmoment (kgmm)
Basic empty mass	1270	3360	4267200	1270	3360	4267200
Pilot	80	1890	151200	1350	3273	4418400
Bulk cargo	320	3900	1248000	1670	3393	5666400 ¹⁾
Fuel, supply tank	74	2275	168350	1744	3346	5834750 ²⁾
Fuel, main tank	382	3220	1230040	2126	3323	7064790 ³⁾
Flight B: 2 pilots (2 x 80 kg), 3 passengers (1 x 80 kg, 2 x 90 kg), and full fuel (456 kg). The resultant CG is solved as follows:						
	Mass (kg)	Arm (mm)	Massmoment (kgmm)	Resultant		
	Mass (kg)	Arm (mm)	Massmoment (kgmm)	Mass (kg)	Arm (mm)	Massmoment (kgmm)
Basic empty mass	1270	3360	4267200	1270	3360	4267200
Pilot and copilot	160	1890	302400	1430	3196	4569600
Passengers (3)	260	2740	712400	1690	3125	5282000 ¹⁾
Fuel, supply tank	74	2275	168350	1764	3090	5450350 ²⁾
Fuel, main tank	382	3220	1230040	2146	3113	6680390 ³⁾
NOTE To make certain that the CG does not "wander" outside of the CG envelope during the flight, the helicopter must be so loaded that the above points ¹⁾ (minimum landing mass), ²⁾ (supply tank full), and ³⁾ (supply and main tanks full) lie within the approved limits.						

NOTE The lateral CG position will always remain within limits regardless of the location of internal loads provided that the gross mass remains within limitations.

6.2 BASIC EMPTY MASS CENTER OF GRAVITY

The procedure for establishing mass and moment (relative to the reference datum) of the empty helicopter is described in the BO 105 Maintenance Manual (MM). The MASS AND BALANCE RECORD (Form MBR-1) appended to this Section is used to maintain a continuous history of changes to the basic “as delivered” helicopter mass and balance data.

6.2.1 Mass and Balance Record

The basic empty mass (BEM) and center of gravity (CG) location are determined through actual weighing carried out by the helicopter manufacturer. This data is then entered on the first line of the MASS AND BALANCE RECORD, Form MBR-1 which then becomes a permanent part of the Flight Manual.

The MASS AND BALANCE RECORD must be updated (normally by transcribing the applicable information from the EQUIPMENT LIST) when necessary as follows:

- When additional equipment is installed on the helicopter necessitating a change in the basic empty mass (as per definition), add the new entry or entries to the previous totals of basic empty mass and moment then compute the new basic empty mass, moment and CG location (arm).
- Likewise, when equipment is removed from the helicopter, subtract the new entry or entries from the previous totals of basic empty mass and moment then compute the new basic empty mass, moment and CG location (arm).

At all times, the last mass and moment entries are considered the current basic empty mass and balance status of the helicopter.

6.2.2 Equipment List

An EQUIPMENT LIST, Form EL-1 is appended to this section and contains optional equipment of the particular helicopter when delivered. Each item on the list is provided with a number and description for identification, together with its mass, arm and moment.

Those items of equipment that were installed when the particular helicopter was initially weighed are so indicated by a check mark (✓) in the “Initial Weighing” column. Therefore the mass, arm and moment of these items are included in the basic empty mass (BEM) data found on the MASS AND BALANCE RECORD, Form MBR-1.